Every Patch in the Dark

Every time the cold dark matter model was redefined to fit data it had failed to predict — a complete, specific record

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The thesis

A scientific model earns trust by sticking its neck out: it names a number in advance, and the universe either confirms it or kills it. By that standard, the cold dark matter model has never once been at risk. In ninety years it has been redefined at least a dozen times, and **not one of those redefinitions was a prediction confirmed.** Every single one was a repair — a new ingredient, a swapped particle, a freed parameter, a weakened assumption — bolted on *after* an observation it had failed to anticipate arrived to embarrass it.

This is the record, change by change, with the specific trigger for each. The pattern is the point. A theory that can absorb any result is not being confirmed by the data; it is being *edited* by the data. What follows is the edit history.

CHANGE 0 — The original inference (1933)

The claim. Fritz Zwicky applied the virial theorem to the Coma cluster and found its galaxies moving far too fast to be held together by the mass he could see. He posited unseen mass — *dunkle Materie* — to make the books balance. (Kelvin had counted “dark bodies” in 1884, Poincaré had named *matière obscure* in 1906, Kapteyn had estimated a local density in 1922, and Oort had claimed a disk excess in 1932 that was *later shown to be a measurement error* — the very first quantitative dark-matter detection was wrong. But Zwicky’s cluster result is the durable origin.)

What it was. Not yet a model — a single accounting discrepancy and one word to cover it. It was then ignored for forty years. Note what was *not* done: the obvious alternative — that the law of gravity is incomplete where it is very weak — was on the table and was not pursued. The field chose invisible matter over new physics at the very first fork, and has defended that choice ever since.

CHANGE 1 — From clusters to galaxy halos (1974)

The trigger. Vera Rubin and Kent Ford (from 1970) found spiral galaxies rotating *flat* far out — edge stars circling as fast as inner ones, impossible if mass follows light. Albert Bosma (1978) traced the same flatness in radio hydrogen well past the visible disk. Independently, Ostriker and Peebles showed a bare disk would be dynamically unstable.

The change. In 1974 two papers (Ostriker, Peebles & Yahil; Einasto, Kaasik & Saar) hardened the loose 1933 idea into a *model*: every galaxy sits inside an extended halo of unseen matter roughly **ten times** its visible mass. Dark matter went from a cluster curiosity to a universal component of every galaxy.

The cost. The model now required invisible matter *everywhere*, in an amount and arrangement nobody could see — set, in each galaxy, by exactly what was needed to flatten that galaxy's curve.

CHANGE 2 — From ordinary matter to a new particle (late 1970s)

The trigger. Big Bang nucleosynthesis matured into a precise calculation, and it pinned the total amount of *baryonic* matter — protons and neutrons — to a value far below what the halos demanded. The dark matter could not be cold gas, dust, dead stars, or rocks.

The change. Dark matter was redefined, by elimination, as a **non-baryonic particle not present in the Standard Model of physics.** The problem was exported from astronomy to particle physics, and a hypothetical new particle was written into cosmology to fill the gap.

The cost. The model now demanded the existence of a particle no experiment had ever seen, and which the established theory of matter did not contain. It has still never been seen.

CHANGE 3 — Hot dark matter adopted (≈1980)

The trigger. A Soviet experiment (Lyubimov, 1980) claimed the neutrino had a mass near 30 electron-volts — enough for the universe's existing neutrinos to *be* the dark matter. A known particle, no new physics required.

The change. The dark matter was identified as the light, fast neutrino — "hot" dark matter — and for a few years this was the favored model.

The cost. None yet visible. It looked, briefly, economical. Hold that thought.

CHANGE 4 — Hot swapped for cold (1982–1984)

The trigger. Fast neutrinos stream out of small clumps before those clumps can collapse, so hot dark matter builds the *largest* structures first and lets galaxies fragment out later — a top-down universe. The galaxy redshift surveys of the early 1980s, and the simulations of White, Frenk and Davis (1983), showed the opposite: small things formed first, bottom-up. Hot dark matter built the wrong universe.

The change. The particle was swapped again — this time by *temperature*. Out went the hot neutrino; in came **cold dark matter (CDM)** (Peebles 1982; Blumenthal, Faber, Primack & Rees 1984): heavy, sluggish, hypothetical particles that clump on every scale and build structure bottom-up. This is the "CDM" that survives, in patched form, to today.

The cost. The economical, known-particle answer (neutrinos) was abandoned. The dark matter was now a *specifically* exotic, cold, never-detected particle, chosen not because it had been found but because it produced the right kind of universe. The candidate had been redefined twice — non-baryonic, then cold — in five years, each time by data, never by discovery.

(In 1983, in the same window, Milgrom proposed Modified Newtonian Dynamics — the road not taken at the 1933 fork: change the law, not the matter. It would go on to fit galaxy rotation curves with a single number while the dark-matter program kept adding them.)

CHANGE 5 — Critical-density "Standard CDM" (mid-1980s)

The trigger. Inflation theory predicted a spatially flat universe. Assuming matter supplies the flatness, the total matter density must equal the critical value: $\Omega_m = 1$.

The change. Cold dark matter at exactly critical density became **Standard CDM (SCDM)** — the establishment model for over a decade, almost all of it dark matter.

The cost. A specific, falsifiable commitment ($\Omega_m = 1$) made not because the matter had been weighed and found to be critical, but because a *theoretical* preference required it.

CHANGE 6 — The cosmological constant bolted on: SCDM → ΛCDM (1998)

The trigger. This is the big one, and it had been screaming for nearly a decade. Galaxy clustering already argued for a low matter density plus a cosmological constant (Efstathiou, Sutherland & Maddox, 1990) — and was set aside as unfashionable. COBE's 1992 CMB map made SCDM over-predict small-scale clumpiness. Standard CDM was, by the mid-1990s, failing on three independent fronts at once: the universe it described came out *younger than its own oldest stars* (the age problem); galaxy clusters held *too much ordinary matter* for $\Omega_m = 1$ (the "baryon catastrophe," White, Navarro, Evrard & Frenk 1993); and the clustering of galaxies implied *less* matter than the model contained. The 1990s response was telling: not one fix but a *bazaar* of them — tilted CDM, mixed hot-plus-cold CDM, open CDM, decaying-particle τCDM. Then, in 1998–99, distant supernovae revealed the expansion is *accelerating*.

The change. A second invisible component — **dark energy**, the cosmological constant Λ — was added. Matter was dropped to $\Omega_m \approx 0.3$, Λ supplied the remaining ≈ 0.7, and the model became **ΛCDM**: about 5% ordinary matter, ~27% cold dark matter, ~68% dark energy. Lowering the matter and adding Λ resolved the age problem, the baryon fraction, and the clustering shape simultaneously — which is exactly why it was adopted.

The cost. Enormous, and rarely stated plainly. The universe now required **two** unseen components, together about **95% of everything**, neither ever detected in a laboratory. And the value of Λ is the single worst quantitative estimate in the history of physics: the natural prediction from quantum theory overshoots the measured value by some **120 orders of magnitude**. ΛCDM did not explain dark energy. It named it, set its value by hand to whatever the supernovae required, and moved on. A model that was Ω_m = 1 in 1995 was Ω_m = 0.3 plus a 120-orders-of-magnitude fudge by 1999 — and this is remembered as a triumph.

On the "triumphs" that followed. It is true that the same six-parameter ΛCDM then fit the detailed acoustic peaks of the cosmic microwave background (WMAP 2003, Planck) and predicted the baryon-acoustic "ruler" later found in galaxy surveys. These are the model's real successes, and they should be understood for exactly what they are: *consistency fits, with the dark sector already assumed.* The CMB does not discover dark matter — it *assumes* a cold, collisionless component and measures **how much** of it you must add to fit the peak heights. The dark-matter density is an input dialed to the data, not an output. It is a genuine achievement that one framework, once you grant it dark matter, dark energy, and inflation, fits the early universe with six knobs. It is not the same thing as having found any of those three ingredients — and, as the next changes show, even this flagship fit has begun to crack from the inside.

CHANGE 7 — Self-interacting dark matter (2000)

The trigger. Higher-resolution simulations exposed cold dark matter failing *inside* galaxies. CDM halos should rise to a dense central "cusp"; real dwarf galaxies show flat "cores" (the cusp–core problem, Flores & Primack and Moore, 1994). CDM predicts hundreds to thousands of satellite subhalos around a galaxy like ours; only dozens are seen (the missing-satellites problem, Klypin et al. and Moore et al., 1999).

The change. Spergel and Steinhardt redefined the particle *again* — now it could collide with itself (**self-interacting dark matter**) — specifically to smear cusps into cores.

The cost. A new free parameter (the self-interaction strength), introduced for the sole purpose of erasing a known failure.

CHANGE 8 — Baryonic feedback tuning (2000s–2010s)

The trigger. The small-scale failures multiplied: "too big to fail" (Boylan-Kolchin, Bullock & Kaplinghat, 2011 — the biggest predicted subhalos are too dense to match the observed satellites); the thin, rotating planes of satellites around the Milky Way and Andromeda; the diversity of rotation-curve shapes. And looming over all of it, the radial acceleration relation (McGaugh, Lelli & Schombert, 2016): across 153 galaxies the gravity is fixed almost entirely by the *visible* matter, through a single universal acceleration — a tight law dark matter has no reason to obey and must be arranged, galaxy by galaxy, to reproduce.

The change. Rather than touch the particle again, the mainstream tuned the *astrophysics*: large simulations (FIRE, EAGLE, IllustrisTNG) added supernova winds and black-hole outflows

— "baryonic feedback" — with enough adjustable strength to soften cusps, dim satellites, and reproduce the counts.

The cost. The model's small-scale predictions became whatever the feedback was tuned to make them. Processes were calibrated *to reproduce the very data they were invoked to explain* — the textbook signature of a fit dressed as an explanation.

CHANGE 9 — The candidate zoo (2000s–2010s)

The trigger. The favored particle — the WIMP — refused to appear, and the small-scale problems persisted.

The change. When one invisible particle would not behave, the field added others to choose from: **warm dark matter** (a keV sterile neutrino, to wipe out excess small structure), **fuzzy / ultralight dark matter** (an axion-like particle of mass ~10^{-22} eV, its quantum wavelength galaxy-sized, to make cores), and a **revival of primordial black holes** as dark matter (re-energized in 2016 after LIGO). Each is a different invisible thing; each is tuned to relieve a different tension.

The cost. "Dark matter" stopped being a hypothesis and became a *category* — a placeholder into which any new particle could be slotted as needed. A hypothesis names one thing; a category absorbs everything.

CHANGE 10 — Quietly retiring the WIMP (2017–2024)

The trigger. Forty years of searching. Underground detectors grew from kilograms to tonnes — DAMA (whose lone claimed signal nobody else could reproduce), CDMS, XENON, LUX, then XENON1T, LUX-ZEPLIN, PandaX — and found **nothing**, excluding the WIMP across nearly all of its natural parameter space, down toward the irreducible neutrino background. The Large Hadron Collider found no supersymmetry and made no WIMPs. Indirect hints (the galactic-center gamma-ray excess, a 3.5 keV line) each flared and faded.

The change. The candidate that *defined* the field for forty years was, without ceremony, demoted — attention shifted to axions, ultralight fields, and primordial black holes — rather than the premise being questioned.

The cost. This is the deepest tell of all. The dark-matter program's central prediction — a weakly interacting massive particle at the electroweak scale — was *tested to destruction and falsified*, and the model did not die. It simply changed which invisible thing it was about. A theory that survives the death of its own flagship prediction is a theory that was never going to be allowed to fail.

CHANGE 11 — Un-fixing the cosmological constant (2024–2025)

The trigger. Two tensions hardened into crises. The present expansion rate from the CMB ($H_0 \approx 67.4$) disagrees with the rate measured locally ($H_0 \approx 73$) at more than **five sigma** — the Hubble tension. Structure-growth surveys (KiDS, DES) find the universe slightly less clumpy than CMB-calibrated ΛCDM predicts — the S_8 tension. Then the Dark Energy Spectroscopic Instrument

(2024–2025) found its baryon-acoustic data *prefer a dark energy that changes with time* ($w \neq -1$) over a constant Λ, at up to ~4σ.

The change. The one number ΛCDM treated as a fixed constant — Λ itself — is being promoted to a time-varying function w(z), turning ΛCDM into w_0w_aCDM, two free functions where there was one constant. For the Hubble tension, an extra component — "early dark energy" — is being engineered to switch on briefly before recombination, raise the inferred rate, and vanish.

The cost. Even the model's signature successes are now being paid for with new parameters. The acoustic "ruler" that confirmed ΛCDM in 2005 is, via the same instrument's newer data, now pulling it toward yet another ingredient. The flagship CMB fit disagrees with the local universe at five sigma. The model is being edited from inside its own best evidence.

CHANGE 12 (pending) — Breaking the universe's symmetry (2021–present)

The trigger. The deepest challenge yet does not touch a parameter; it touches the *foundation*. ΛCDM assumes the **cosmological principle**: on large scales the universe looks the same in every direction. The clean test: our motion should tip the distant sky into a "dipole" matching the one the CMB shows (our speed ≈ 370 km/s), in any population of distant sources, with no freedom. Nathan Secrest and collaborators measured it. Using 1.36 million WISE quasars (2021), the dipole came out **more than twice** the predicted amplitude — **4.9σ**. Adding independent NVSS radio galaxies (2022), **5.1σ**, under a paper titled, flatly, *A Challenge to the Standard Cosmological Model*. Either we move twice as fast as the CMB says — contradicting the CMB — or the distant universe is genuinely not isotropic, and the assumption the entire model rests on is wrong.

The change (anticipated). If the dipole holds, the response will follow the pattern: a controlled departure from isotropy — a "tilted" universe, a giant-scale structure, an anisotropic mode — added with just enough freedom to fit the observed dipole while preserving the CMB's near-perfect isotropy. A new ingredient, again, this time at the level of the founding symmetry.

Honest status. The anomaly is real, independent across catalogs, and persistent — and it is under active reassessment (Bayesian estimators, reworked CatWISE samples, kinematic source-count corrections, 2023–2025). It is not a settled overthrow. It is the sharpest open test of whether *any* measurement exists that this model cannot absorb. On the record of the eleven changes above, the safe bet is that it will be absorbed.

The ledger

Count them. The cold dark matter model has been redefined, specifically and on the record:

1. cluster mass ⟶ universal galaxy halos (1974);
2. ordinary matter ⟶ a non-baryonic new particle (late 1970s);
3. ⟶ hot (neutrino) dark matter (≈1980);
4. ⟶ cold dark matter (1982–84);
5. ⟶ critical-density Standard CDM (mid-1980s);

6. ⟶ ΛCDM, a second invisible component bolted on (1998);
7. ⟶ self-interacting dark matter (2000);
8. ⟶ feedback-tuned to fit small scales (2000s–2010s);
9. ⟶ a zoo of interchangeable candidates (2000s–2010s);
10. ⟶ the flagship WIMP quietly retired after non-detection (2017–2024);
11. ⟶ the cosmological constant promoted to a time-varying field (2024–2025);
12. ⟶ (pending) the cosmological principle itself weakened, if the dipole holds.

Not one of these twelve was a number named in advance and then confirmed. Every one was a repair forced by data the model had not foreseen — a particle swapped, a component added, a parameter freed, a foundation softened. The model's defining property is not predictive power. It is *adjustability*: an ability to fit Neptune and Vulcan alike, to greet every new datum as either a confirmation or a free parameter, and never as a refutation.

That is the real indictment, and it needs no exaggeration. A theory you cannot kill is not a strong theory. It is an unfalsifiable one wearing the costume of a strong one. The honest measure of a cosmology is not how much it can fit after the fact — by that measure ΛCDM is unbeatable, because it has been built, over ninety years and a dozen edits, to fit anything. The honest measure is whether it ever told you, in advance, a single number it was willing to die for. Cold dark matter, in ninety years, never has.

A real theory of the missing gravity would do the opposite. It would name the scale — fix it, once, from something already known, with no freedom to adjust — and then stand or fall on whether that one number is right. It would make the galactic acceleration scale a *consequence*, not a fit; it would predict how that scale changes with cosmic time and dare you to measure it; it would pass or fail a solar-system test cleanly. Such a theory can be killed by a single observation, which is precisely why it is worth more than one that has spent a century learning how to survive them all.

The cosmic dipole is the latest knock on the door. The history says the door will be answered, as it always has been, with one more invisible patch. The question worth asking is how many patches a model is allowed before we admit it was the wrong picture from the start — and whether the answer the galaxies have been pointing at the whole time was not *more dark stuff*, but a different law.

Sources

The standard scholarly history is Bertone & Hooper, *A History of Dark Matter*, Rev. Mod. Phys. **90**, 045002 (2018) [arXiv:1605.04909], which documents the early inferences and the hot-to-cold transition. Key primary points: Zwicky (1933); Rubin & Ford (1970), Bosma (1978); Ostriker, Peebles & Yahil (1974) and Einasto, Kaasik & Saar (1974); White, Frenk & Davis (1983) and Blumenthal, Faber, Primack & Rees (1984) on hot⟶cold; Efstathiou, Sutherland & Maddox (1990) and White, Navarro, Evrard & Frenk (1993) on the road to Λ; Riess et al. (1998) and Perlmutter et al. (1999) on acceleration; Spergel & Steinhardt (2000) on SIDM; Boylan-Kolchin et al. (2011) and McGaugh, Lelli & Schombert (2016) on the small-scale and acceleration-relation failures; the LZ/XENON/PandaX exclusion limits (2022–2024); DESI (2024–2025) on evolving dark energy; and Secrest et al. (2021, *ApJL* 908 L51; 2022, *ApJL* 937 L31) on the cosmic dipole, with the 2023–2025 reassessment literature.

Every dated change above is part of the documented historical record; the framing is the author's. The model's genuine consistency fits (the CMB acoustic peaks, the baryon-acoustic ruler) are described accurately as fits with an assumed dark sector, not denied.